EV Technology Course — Module 17: Capstone — EV Teardown and Should-Cost Project

Oct 14, 2026 · @Ashishpreet

This capstone brings the whole course together. You will plan and carry out (on paper or with real parts) a teardown of a compact electric SUV, map every system you studied in Modules 1–16, build a vehicle-level should-cost, benchmark it, and propose VA/VE and technology upgrades. This is the exact workflow used by OEM cost-engineering and benchmarking teams.

1. Learning objectives

- Plan a structured vehicle teardown with safety, documentation and data capture.
- Build a system-level BOM for an EV and identify materials and processes.
- Build a bottom-up should-cost for key EV systems and roll it up to vehicle level.
- Benchmark against competitor vehicles and identify cost and technology gaps.
- Write a concise VA/VE and technology recommendation report for management.

2. Fundamentals

- **Teardown** = disassembling a product to understand design, materials, processes and cost.
- **Benchmarking** = comparing your product to competitors on cost, mass, performance, features and manufacturability.
- **Should-cost roll-up** = part should-costs → subsystem → system → vehicle.
- **Output** is not only a number but a list of actionable ideas with savings, investment and risk.
- **Safety first:** EV teardowns involve high voltage, stored energy in batteries and capacitors, and hazardous materials.

3. Detailed technical explanation

Project scope (example)

- Vehicle: a compact electric SUV with ~40 kWh LFP pack, front e-axle (~100 kW), 7.2 kW AC and ~50 kW DC charging.

- Deliverables: teardown database, system BOM with masses, should-cost model, benchmark sheet, VA/VE idea list, final report and presentation.

Teardown phases

Phase	Activities	Modules used
1. Plan and safety	HV de-energisation, isolation checks, PPE, discharge of DC link, battery storage plan	6, 12
2. Vehicle-level data	Mass, axle loads, dimensions, range test, charging test, NVH, photos	1, 11
3. Systems removal	Battery pack, e-axle, PEU/OBC/DC-DC, thermal system, HV harness, LV ECUs	2–7, 10
4. Body and chassis	BIW, closures, suspension, brakes, steering, seats, interior, glass	8, 9
5. Part-level analysis	Weigh, measure, identify material (XRF, FTIR), process, supplier marks	13
6. Costing	Should-cost per part, roll-up, sensitivity	13
7. Benchmark and ideas	Compare with competitors, VA/VE, tech roadmap	14–16

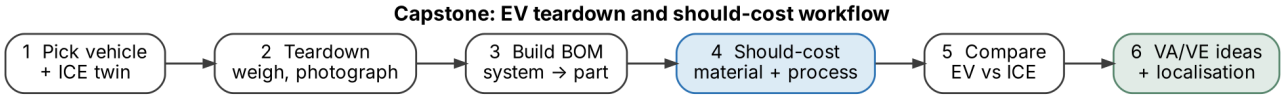
Data to capture per part

- Part name and number, system, quantity, mass, material, process, dimensions, supplier, photos, joining method, special features, estimated cycle time, tooling type.

Costing approach by commodity

- Plastics: material × gross weight + MHR × cycle time ÷ cavities + paint/assembly.
- Sheet metal: blank weight × rate − scrap value + press strokes × rate.
- Castings: poured weight × alloy rate + casting conversion + machining.
- Electronics: PCB area and layers, component BOM (from a parts database), SMT and test cost.
- Battery: cells (₹/kWh or ₹/cell) + structure + BMS + thermal + assembly.
- Bought-outs (tyres, glass, seats): benchmark prices.

4. Diagram: capstone workflow



5. Step-by-step: running the project

1. Define objectives and the comparison set (your vehicle plus 2–3 competitors).
2. Prepare the safety plan and teardown area; train the team on HV safety.
3. Record vehicle-level tests (range, charging curve, mass) before disassembly.
4. Disassemble system by system; photograph, label and log every part.
5. Build the BOM in a spreadsheet: system → subsystem → part, with mass, material and process.
6. Should-cost the top 80% of value parts in detail (battery, e-drive, PEU, BIW, seats, thermal, ECUs); use benchmarks for the rest.
7. Roll up to vehicle cost; add assembly, warranty, overheads.
8. Benchmark against competitors: cost, mass, Wh/km, features.
9. Generate VA/VE and technology ideas with savings, investment, risk and timing.
10. Present a one-page summary and backup sheets to management.

6. Engineering equations

Vehicle should-cost roll-up:

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C_{vehicle} = \sum_{systems} C_{system} + C_{assembly} + C_{warranty} + C_{overhead}
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Cost share of a system:

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\text{Share}_i = \frac{C_i}{C_{vehicle}} \times 100\%
```

Idea value (net present saving, simple):

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S_{net} = (\Delta C_{unit} \times V_{life}) - I_{capex+validation}
```

7. Worked examples

Example 1: illustrative vehicle roll-up (₹, material + conversion per vehicle).

System	Cost (₹)	Share
Battery pack (40 kWh × ₹9,000)	3,60,000	33.7%
BIW and closures	1,10,000	10.3%
Chassis (suspension, brakes, steering, wheels, tyres)	95,000	8.9%
E-drive (Module 13 should-cost)	89,400	8.4%
Interior and seats	90,000	8.4%
LV electrical, ECUs, infotainment	85,000	8.0%
Exterior (bumpers, lamps, glass)	60,000	5.6%
Thermal system	45,000	4.2%
Power electronics (OBC, DC-DC, PDU)	35,000	3.3%
Final assembly and EOL	30,000	2.8%
Paint	25,000	2.3%
ADAS sensors and controller	25,000	2.3%
HV harness	18,000	1.7%
Total	**10,67,400**	**100%**

Adding warranty (3%) and overheads (8%) gives about **₹11.85 lakh** before OEM margin, dealer margin and taxes.

Example 2: impact of battery price. A 12% fall in cell/pack price saves 3,60,000 × 0.12 = **₹43,200** per vehicle, more than the whole power electronics system.

Example 3: idea value. An HV harness re-route saves ₹1,200 per vehicle over 3 lakh lifetime volume (₹36 crore) and needs ₹4 crore of tooling and validation: net saving **₹32 crore**.

8. Real-world examples

- **Munro & Associates** teardowns of Tesla Model 3/Y and Cybertruck shaped how the industry views giga-casting and structural packs.
- **UBS teardown of the Chevrolet Bolt and BYD Seal** highlighted BYD's vertical-integration cost advantage.

- **A2Mac1 and Caresoft** provide benchmarking databases used by Indian OEMs and suppliers.
- Indian OEMs routinely benchmark imported EVs (e.g. BYD, MG, Hyundai) before localising platforms.

9. Manufacturing applications

- Teardown data feeds design-to-cost targets for the next platform.
- Localisation planning: which imported parts (cells, SiC, magnets, ECUs) can be sourced locally and when.
- Supplier negotiation: cost-element-wise should-cost sheets.
- Process benchmarking: competitor joining methods (adhesives, rivets, laser welds), casting vs stamping, cell-to-pack designs.
- Quality lessons: failure-prone designs seen in teardowns guide DFMEA.

10. Common mistakes and misconceptions

- **Skipping HV safety steps:** batteries and capacitors can hold lethal energy even when the car is "off".
- **Costing every small part in detail:** focus on the top 20% of parts that hold 80% of cost.
- **Ignoring vehicle-level effects:** a cheaper part that adds mass or Wh/km may cost more in battery.
- **Presenting numbers without confidence ranges:** state assumptions and sensitivity (commodity prices, volume).

11. Practical case study: your capstone brief

Brief. Management wants the next compact EV SUV to cost 10% less than the current model, with the same real-world range of 300 km.

Approach. Use the teardown roll-up to find big levers: LFP cell-to-pack (−₹25,000–40,000), efficiency improvements to shrink the pack by 3 kWh (−₹27,000), IGBT for base variant (−₹7,000), zonal harness (−₹6,000), seat and interior part consolidation (−₹8,000), local magnets and cells (−₹10,000–20,000).

Your task. Build the idea list with savings, investment, timing and risk, and show whether the 10% target (~₹1.2 lakh) is achievable.

Lesson. EV cost reduction is a whole-vehicle exercise: battery, efficiency and integration work together.

12. Revision notes

- Plan → safety → vehicle tests → teardown → BOM → should-cost → roll-up → benchmark → ideas → report.
- Battery is about one-third of cost; it is the biggest lever.
- Focus detailed costing on high-value parts; benchmark the rest.
- Ideas need savings, investment, risk and timing to be decision-ready.

13. Glossary

Term	Meaning
Teardown	Systematic disassembly for analysis
Benchmarking	Comparison with competitors
Roll-up	Aggregating part costs into system and vehicle cost
Design-to-cost	Setting cost targets that guide design
XRF / FTIR	Methods to identify metals and polymers
DFMEA	Design failure mode and effects analysis

14. Knowledge check

Conceptual

1. Why record vehicle-level tests before teardown?
2. Which EV systems deserve the most detailed costing?
3. What HV safety steps come before removing the battery?
4. Why include Wh/km in a cost benchmark?
5. What makes a VA/VE idea "decision-ready"?

Numerical

1. A motor uses 1.2 kg of magnets at ₹6,500/kg. Magnet cost?
2. An aluminium housing of 10.5 kg at ₹330/kg. Material cost?
3. A competitor's 52 kWh pack at ₹8,500/kWh. Pack cost?

Application

1. Draft the column headings of your teardown database.

2. Outline the one-page management summary of your capstone.

15. Answer key

Conceptual

1. Range, charging, mass and performance data are lost once the vehicle is disassembled.
2. Battery pack, e-drive, power electronics, BIW, seats and interior, thermal system, ECUs and infotainment.
3. Disconnect the 12 V battery, remove the service disconnect, wait for capacitor discharge, verify zero voltage with a rated meter, check isolation, use HV PPE and insulated tools.
4. Efficiency changes battery size and therefore cost; a cheaper but less efficient part may cost more overall.
5. Quantified saving per unit, investment, payback, validation needs, risk, timing and owner.

Numerical

1. 1.2 × 6,500 = **₹7,800**.
2. 10.5 × 330 = **₹3,465**.
3. 52 × 8,500 = **₹4,42,000**.

Application

1. Part ID, system, subsystem, part name, quantity, mass (g), material, process, dimensions, supplier/marking, photo link, joining method, cycle time estimate, tooling type, should-cost (material, conversion, overhead, profit), benchmark cost, remarks/ideas.
2. Objective and vehicle; key findings (cost by system, vs competitors); top 10 ideas with ₹ savings, capex and timing; total achievable saving vs target; risks and decisions needed; next steps.

16. Further learning

- A2Mac1 and Caresoft benchmarking resources.
- Munro & Associates teardown videos.
- Your own Battery course Modules 6, 12 and 15 for pack design, manufacturing and industry data.
- Revisit Modules 1–16 of this course for each system's details.

Congratulations — you have completed the EV Technology Course.